

Simulation of a single-server queuing system: (1.4)

Question: $n=6$, Arrivals occur at times 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8 and 7.2. Departures (service completions) occur at times 2.4, 3.1, 3.3, 4.9 and 8.6 and the simulation ends at time $T(6) = 8.6$.

Soln: Arrivals: 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8, 7.2

Departures: 2.4, 3.1, 3.3, 4.9, 8.6

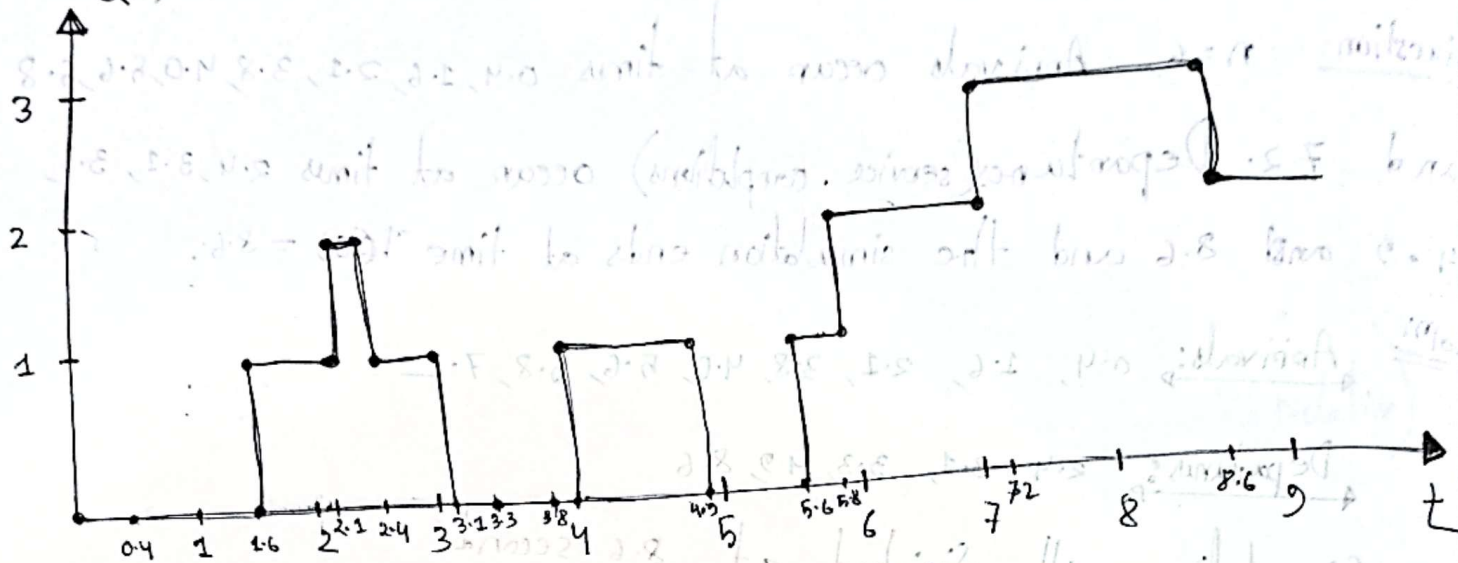
$\therefore$ Simulation will finished at 8.6 second

Initially, Time = 0, Queue = 0, Server = Idle

0.4 (arrivals) $Q=0$	1.6 (arrivals) $Q=1$	2.1 (arrivals) $Q=2$	2.4 (departures) $Q=1$	3.1 (departures) $Q=0$
3.3 (departures) $Q=0$	3.8 (arrivals) $Q=1$	4.0 (arrivals) $Q=2$	4.9 (departures) $Q=0$	5.6 (arrivals) $Q=1$
5.8 (arrivals) $Q=2$	7.2 (arrivals) $Q=3$	8.6 (departures) $Q=2$		

Interval	0 - 0.4	0.4 - 1.6	1.6 - 2.1	2.1 - 2.4	2.4 - 3.1	3.1 - 3.3	3.3 - 3.8	3.8 - 4.0	4.0 - 4.9	4.9 - 5.6
Q_t	0	1	2	2	1	0	0	0	1	1
Interval	5.6 - 5.8	5.8 - 7.2	7.2 - 8.6							
Q_c	1	2	3							

Graph: $Q(t)$



expected average number of customers, $\hat{q}(n) = \sum_{i=0}^3 i T_i$

$Q(t) = 0$, Interval: $(0-0.4)$, $(0.4-1.6)$, $(3.1-3.3)$, $(3.3-3.8)$, $(3.8-4.0)$, $(4.9-5.6)$

$$T_0 = (0.4-0) + (1.6-0.4) + (3.3-3.1) + (3.8-3.3) + (4.0-3.8) + (5.6-4.9) = 3.2$$

Similarly,

$$T_1 = (2.1-1.6) + (3.1-2.4) + (4.9-4.0) + (5.8-5.6) = 2.3$$

$$T_2 = (2.4-2.1) + (7.2-5.8) = 1.7$$

$$T_3 = (8.6-7.2) = 1.4$$

$$\hat{q}(n) = (0 \times 3.2) + (1 \times 2.3) + (2 \times 1.7) + (3 \times 1.4) = 9.9$$

total simulation time, $T(n) = 8.6$

$$\text{Average, } \hat{q}(n) = \frac{9.9}{8.6} = 1.15$$

Means, through the whole simulation, average queue length about 1.15

Server utilization: $\hat{u}(n)$

$$\text{Define by, } B(t) = \begin{cases} 1 & \text{server busy} \\ 0 & \text{server idle} \end{cases}$$

$$\hat{u}(n) = \frac{\text{server busy}}{\text{total time}}$$

Ideal time : $(0 - 0.4)$ and $(3.3 - 3.8)$

$$\text{Busy} = \text{Total} - \text{ideal} = 8.6 - (0.4 + (3.8 - 3.3)) = 7.7$$

$$\hat{u}(n) = \frac{7.7}{8.6} = 0.895 \approx 90$$

Means, server busy about 90% time of the simulation.

Given,

Arrivals: 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8, 7.2

Interarrival time

$$A_1 = 0.4 - 0 = 0.4$$

$$A_2 = 1.6 - 0.4 = 1.2$$

$$A_3 = 3.8 - 2.1 = 1.7$$

$$A_4 = 2.1 - 1.6 = 0.5$$

$$A_5 = 3.8 - 2.1 = 1.7$$

$$A_6 = 4.0 - 3.8 = 0.2$$

$$A_7 = 5.6 - 4.0 = 1.6$$

$$A_8 = 5.8 - 5.6 = 0.2$$

$$A_9 = 7.2 - 5.8 = 1.4$$

De Service time

Arrivals: 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8, 7.2

Departures: 2.4, 3.1, 3.3, 4.9, 8.6

$$S = (\text{Departure} - \text{Arrivals})$$

$$S_1 = (2.4 - 0.4) = 2$$

$$S_2 = (3.1 - 2.4) = 0.7$$

$$S_3 = (3.3 - 3.1) = 0.2$$

$$S_4 = (4.9 - 3.8) = 1.1$$

$$S_5 = 8.6 - 4.9 = 3.7$$

(3.3 - 3.8 serve idle for)

example:02 : আমি doctor এর কাছে যাবো। আমি মাথায় ব্যাণ্ডেজ ২ জন line এ দাঁড়িয়ে আছি।
 প্রথম জন ৩s আমি ৫s এ চলেছি। প্রথম লোক ৩s এ doctor এর কাছে যায় আর
 বের হয়ে যায় ১২s আমি নিয়ে বের হই ১৬ সেকেন্ডে। Interarrival time,
 service time, Delay বের করি।

Soln:

Interarrival time, $A_1 = 3s$
 $A_2 = 5 - 3 = 2s$

Service time, $S_1 = 12 - 9 = 3s$
 $S_2 = 16 - 12 = 4s$

Delay, $D_1 = 9 - 3 = 6s$

$D_2 = 12 - 5 = 7s$

$D_{i+1} = \max(D_i + S_i - A_{i+1}, 0)$

$D_2 = \max(D_1 + S_1 - A_2, 0)$

$= \max(6 + 3 - 2, 0)$

$= \max(7, 0)$

$= 7s$

[This formula is given simulation output data & stochastic process ch-04]

→ D_i and D_{i+1} positively correlated (why?)

$D_{i+1} = \max(D_i + S_i - A_{i+1}, 0)$

[D_i বড় হলে, D_{i+1} স্কল ও বড় হওয়ার possibility বেশি, তাই positively correlated.]